Fast answers: indexes

Data engineering · Level 3 · Lesson 4



In this lesson you will

- Explain why reading every row gets slow.
- Count the looks of a linear and a binary search in Python.
- Decide which columns to index.

Key ideas

- Row by row: half the table on average.
- Halving: about 20 looks for a million.
- Indexes trade space for speed.

Guess my number

1. Pairs (2 min): pick a number
2. Guess (5 min): one by one
3. Halve (5 min): always the middle
4. Compare (4 min): why seven?

Let pupils keep score on paper; the record of guesses shows the pattern.

Let’s discuss

1. Why does a linear search get ten times slower on ten times more rows?
2. Why does halving need the data to be sorted?
3. What does an index cost?

Exit ticket

- Write how many looks halving needs for 1,000 rows, and why.